TCA Report - 1

1. Steps Performed

a. Data Loading and Preprocessing

- Imported the trading dataset (project2_trading.csv) and instrument metadata (project2_instruments.csv).
- Computed additional cost metrics:

$$\text{Slippage in USD} = \frac{\text{slippage_bps}}{10000} \times \text{execution_price} \times \text{quantity}$$

$$\text{Market Impact in USD} = \frac{\text{market_impact_bps}}{10000} \times \text{execution_price} \times \text{quantity}$$

$$\text{Total Trading Cost in USD} = \text{slippage_usd} + \text{commission_usd}$$

b. Merging with Instrument Data

- Merged with instrument metadata to bring in **asset class** and **sector**, allowing cost analysis by category.

c. TCA-Focused Dataset Creation

- Extracted relevant columns: timestamp, strategy, asset_class, sector, slippage (bps & USD), market impact, commissions, total trading cost, realized and unrealized PNL.
- This streamlined dataset formed the basis for further analysis.

d. Descriptive Statistics

- Generated statistical summaries (mean, min, max, quartiles) for all TCA metrics.

2. Sector-Level Insights

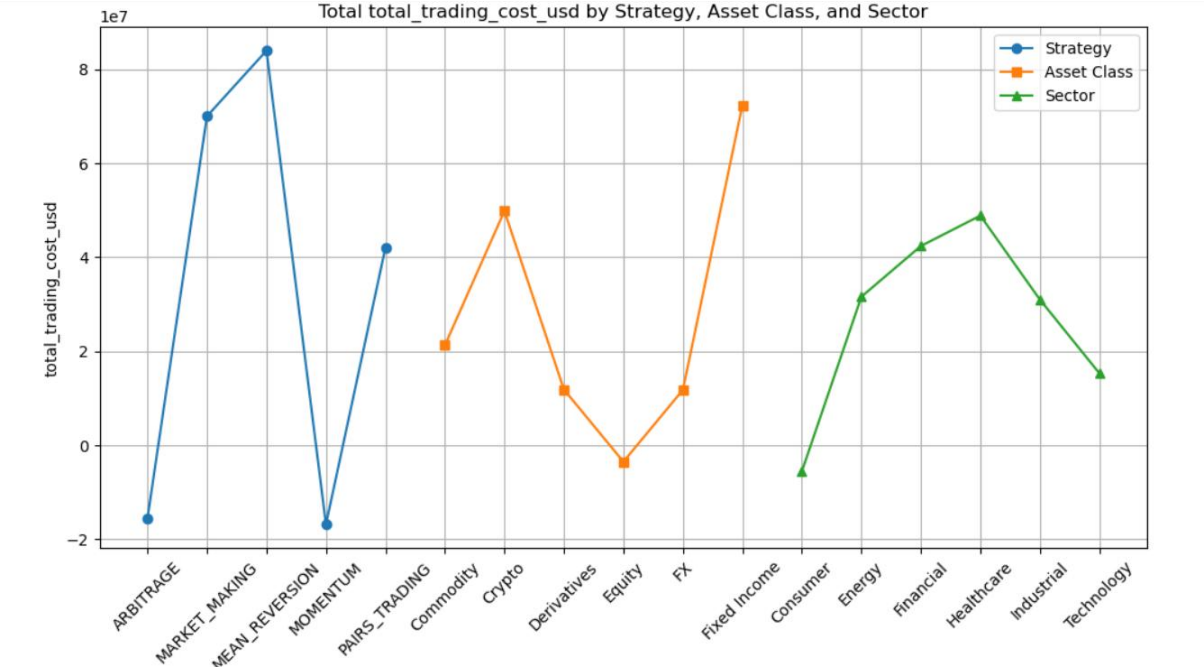
- **Best Sector**:
  Technology – Delivered the highest **average P&L** with relatively low total trading costs. Efficient execution in liquid, high-volume instruments.
- **High-Cost Sector**:
  Healthcare – Exhibited the highest **slippage and total trading costs**, indicating challenges in execution due to lower liquidity or volatile conditions.
- **Balanced Sector**:
  Financial – Showed moderate costs and stable P&L, suggesting reliable but not outstanding performance.

3. Asset Class-Level Insights

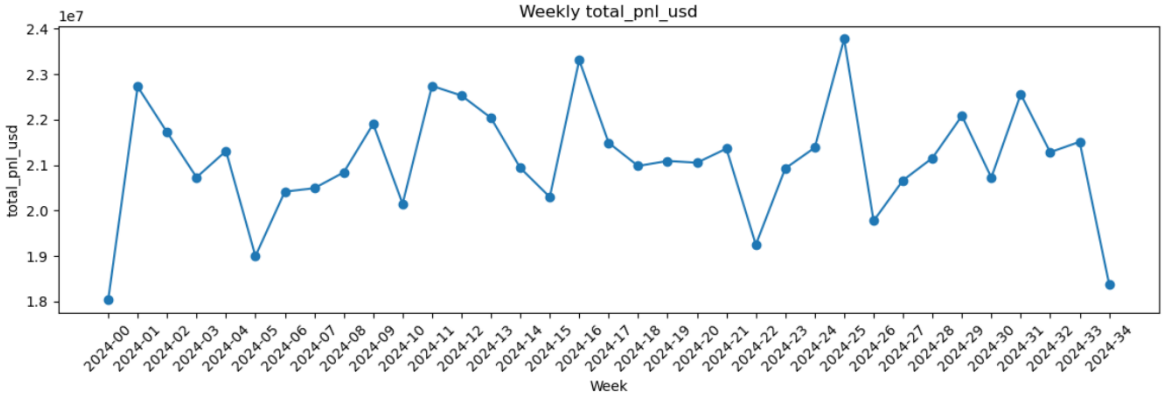
- **Best Performing**:
  Derivatives – Highest profitability with relatively low trading costs. Strong liquidity helped minimize slippage and impact.

- **Costly Class**:
  Fixed Income – Notable for high market impact and higher overall costs, consistent with bond market illiquidity.

- **Mixed Performance**:
  Crypto – Higher slippage, but select trades offered good opportunities, reflecting sensitivity to contract type and market timing.






Weekly total_pnl_usd
1e7
2.4
2.3
2.2
2.1
2.0
1.9
1.8
total_pnl_usd
2024-00
2024-01
2024-02
2024-03
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2024-05
2024-06
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2024-33
2024-34
Week